Statistical Physics for Machine Learning
Exercise 6

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Abstract

We implement the Hebb rule and the Perceptron rule for a $N = 40$ parameters perceptron and estimate the training and test error over many different iterations (with different datasets and teachers). We then compare the results with the Gibbs predictions obtained in the previous exercises.

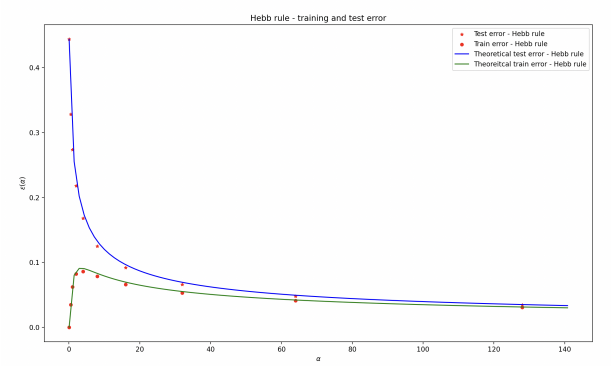

Figure 1: *Average training and test errors obtained, together with the theoretical calculation, for the Hebb rule*

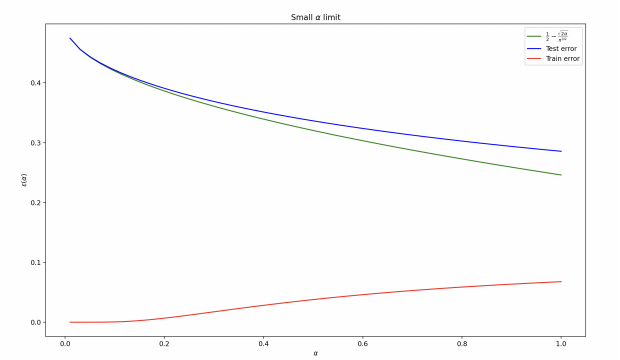

Figure 2: *Small α behavior for the Hebb rule, together with the small α limit of the generalization error $\epsilon(\alpha)$*

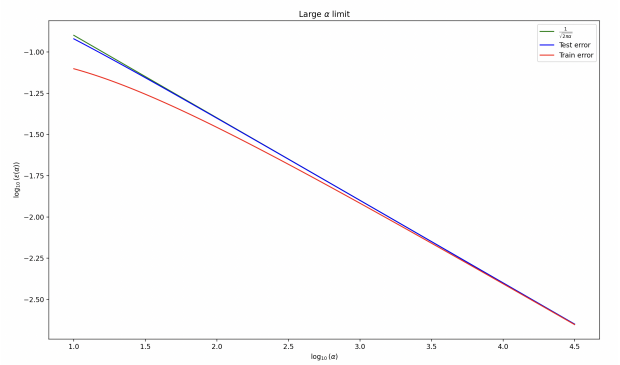

Figure 3: *log-log plot of the large α behavior for the Hebb rule, together with the large α limit of the generalization error $\epsilon(\alpha)$*

Hebb rule

We implement the Hebb learning rule on a $N = 40$ parameters perceptron. We then choose some values of $\alpha = \frac{N}{P}$ and for each of those values we simulate 1000 learning processes with random teachers. We then evaluate the average test and training error over these repetitions, and compare them with their theoretical values. The relevant plot is shown in Fig.(1)

We immediately see that the predictions match very well with the experimental data. Furthermore, we clearly see a different behaviour of the training and test errors. The test error goes down monotonically with the number of examples shown to the perceptron, as it is expected, while the training error has a first initial spike, and then a much more gradual descent. The spike in the training error is due to the very high overfitting capability of the Hebb rule when α is very small.

We investigate the small α and large α limit separately. For the test error, for which we have an easy analytical form, we can find the small α limit via a simple Taylor expansion, and we find

$$\epsilon(\alpha) \sim \frac{1}{2} - \frac{\sqrt{2\alpha}}{\pi^{3/2}}$$

which is a very rapidly decreasing function. This means that the Hebb rule is able to learn the patterns in the data very quickly at small α. Unfortunately, this does not continue to hold for larger α. This is confirmed by performing an expansion in $\frac{1}{\alpha}$ of the generalization error at large α, so that we find

$$\epsilon(\alpha) \sim \frac{1}{\sqrt{2\pi\alpha}}$$

So the learning rate is actually quite slow compared to other typical learning rules, where $\epsilon(\alpha) \sim \frac{1}{\alpha}$. We show these behaviors in Fig.(2) and in Fig.(3)

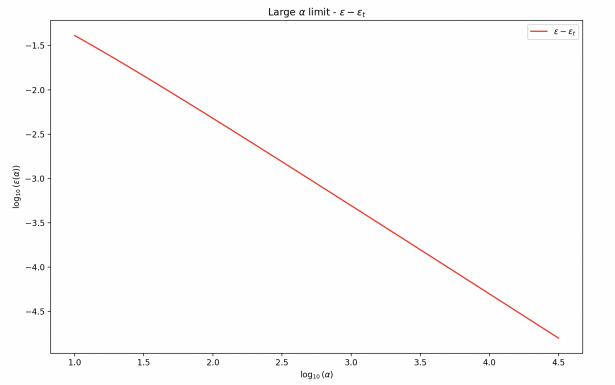

Figure 4: *log-log plot of the difference between the generalization error and test error for the Hebb rule at large α*

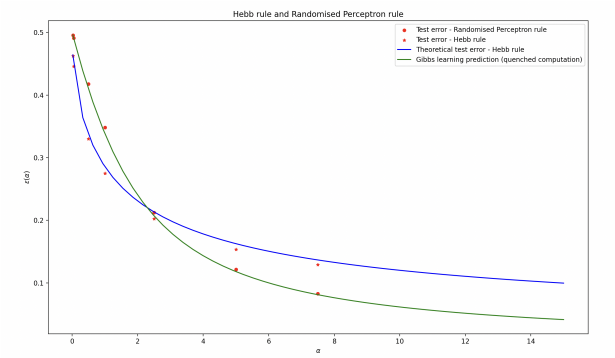

Figure 5: *Experimental results and theoretical computation for the generalization error of the Hebb rule and the Randomised Perceptron rule (Gibbs learning prediction - quenched computation)*

In the large α limit, we see that the training error and the test error seem to converge. To study this property further, we plot the difference between the test and training error as a function of α in Fig.(4)

We see that in the log-log plot, the difference is almost a straight line with coefficient -1. This means that, at large α, the difference between the test and training error goes roughly like

$$(\epsilon - \epsilon_t)(\alpha) \sim \frac{1}{\alpha}$$

Now we superimpose the results obtained so far with the Hebb rule with the results of the Randomised Perceptron rule obtained in previous exercises. The plot is shown in Fig.(5)

As we can see, at first the Hebb rule is much better: it is able to learn very fast and so its generalization error is considerably lower. From $\alpha \sim 2$ on, however, the 'large' α behaviour kicks in: the random perceptron rule is able to have a much lower generalization error due to the scaling $\epsilon \sim 1/\alpha$ instead of $\epsilon \sim 1/\sqrt{\alpha}$ typical of the Hebb rule.

In the last part, we apply the Perceptron rule, that is we don't add the random noise in the training procedure as we did in the randomised perceptron rule of the previous exercises. We simulate 1000 learning scenarios with different teachers and we obtain the results shown in Fig.(6) together with the results of the randomised perceptron rule.

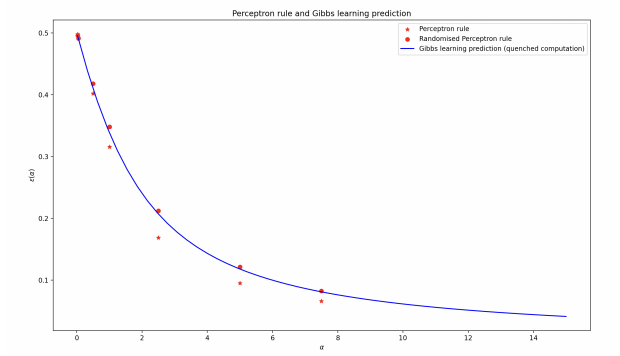

Figure 6: *Experimental results for the Perceptron Rule and Randomised Perceptron rule and Gibbs learning prediction (quenched computation)*

We see that the perceptron rule does not follow precisely the Gibbs learning prediction, while the randomised perceptron rule does. That is because the Gibbs learning prediction is a thermodynamical method that exploits the fact that the learning process gradually explores all of phase space. For the randomised perceptron rule, this is assured by the random noise that we artificially injected inside the learning process. Since the perceptron rule doesn't have this noisy term, it is not well described via a thermodynamical approach.